

MECHANICS PRIME

Before every great machine was a working machine, it was a bad idea that someone refused to abandon. Mechanics Prime puts you in that messy space between an idea and its success. Delegates will build things that may not work, question things that do, and discover that the most useful answer is sometimes another question. There are laws of physics. There are limits of materials. The rest is up to you.

Round 1: Sling, Spot, Solve

ROUND DURATION: 2.5 HOURS

DELEGATE CAP: 2

ELIMINATION: NOT ACTIVE (SUBJECT TO CHANGE TO 50%)

With great power comes great responsibility. With great aim comes... a question. This starts with delegates getting their hands dirty. Delegates will be designing their very own Spider-Man inspired web paper launcher, but this is only the beginning. First, they build their launcher. Then, they find out what it's really capable of. Behind every target is a challenge waiting to be earned. But solving it will take more than good aim. Somewhere in the chaos, things won't quite add up. A flawed system, a failed experiment, a wrong result. Something is hiding in plain sight. Their job is to catch it, figure out what went wrong, and engineer a better way forward.

Round 2: ONCE UPON A DESIGN

ROUND DURATION: 2.5 HOURS (SUBJECT TO CHANGE)

DELEGATE CAP: 2

ELIMINATION: ACTIVE IN THIS ROUND

How much can you improve an invention that was already considered revolutionary? In this round, delegates will take on a historic invention or apparatus and examine it from a modern engineering perspective. They'll need to understand the original design, think critically about its limitations, and use

their creativity and innovation to develop a more practical and effective version. They are required to make a presentation on the spot and present exactly what their changes look like through a properly labelled blueprint, supported by calculations and the concepts behind their decisions. But they won't be the only team working on it. Another team will be tackling the same invention, giving them the chance to put their ideas to the test against their opponents as both teams take turns questioning each other. Delegates will be judged upon: Historic Flaw Analysis, Modern Redesign and Blueprint, Physics and Mechanics Justification, Head-to-Head Pitch and Defense and Peer Critic and Rebuttal. Teams will be tested on May the odds be ever in their favor.

Round 3: THE ODYSSEY

ROUND DURATION:

DELEGATE CAP: 3

Take the current as it comes. Top 5 teams have made it this far—now leave the pages behind. Their task forces are summoned to conquer the aquatic expanse of the pool! Utilizing rationed paperboard shields and sealant films, their squad must architect a full-scale vessel engineered to float, support a human pilot, and traverse the liquid basin under manual propulsion faster than their peers. Hydrostatic equilibrium, volumetric displacement, and material degradation are theirs to balance, yet a single rupture guarantees immediate submersion. Boats will be judged on both engineering and performance: buoyancy, structural integrity, stability, hull shape, paddle design, craftsmanship, and how well a team can explain the physics behind their choices plus how the boat actually does in the water. Finishing position (by recorded time) sets their race score, but only if they stay in the boat and paddle the whole way; leaving to swim or push it, or using any non-hand-paddled propulsion, scores zero on that part regardless of time. Build Wisely. Test boldly. Cross the liquid void, maintain hull stability. Ithaca awaits. Fair winds and steady hands.